INCH-POUND

MIL-P-18948B(OS)
15 July 1992
SUPERSEDING
MIL-P-18948A(OS)
10 November 1972
and Amendment 1
10 May 1974

MILITARY SPECIFICATION

PROJECTILES, PRIMING, MARKING, AND GREASING OF

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers that priming, marking, greasing, and preservation done by the manufacturers of the projectile bodies and related inert components for ammunition calibers 75mm through 16 inch. This specification is supplementary to the specification governing manufacture of projectiles under contract as listed in the applicable list of drawings and specifications (see 6.2).

1.2 Classification. The projectiles shall have one of the following grades of protective coatings as specified in the contract or purchase order (see 6.2):

Grade A - Adherent crystalline phosphate deposit and primer
Grade B - Rust preventive compound

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, 101 Strauss Avenue, Standardization Branch (0343), Indian Head, Maryland 20640-5035, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

TT-C-490	Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings
TT-P-664	Primer Coating, Synthetic, Rust Inhibiting, Lacquer-Resisting

MILITARY

MIL-C-450	Coating Compound, Bituminous, Solvent Type, Black (For Ammunition)
MIL-P-15024	Plates, Tags and Bands for Identification of Equipment
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold Application
MIL-C-18487	Compound, Gun Slushing
MIL-P-22332	Paint, Aircraft, General Purpose, Wide Temperature Range
MIL-G-81322	Grease, Aircraft, General Purpose, Wide Temperature Range, NATO Code Number G-395

STANDARDS

MILITARY

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-650	Explosive Sampling Inspection and Testing
MIL-STD-1168	Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government publications. The following other Government publication forms a part of this document to the extent specified herein. Unless otherwise specified, the issue is that cited in the solicitation.

COMMERCIAL ITEM DESCRIPTION

A-A-1830	Tape, Pressure-Sensitive Adhesive, Box Closure

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to: Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 Standard Test Method of Salt Spray (Fog) Testing (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(NonGovernment standards and other publications are normally available from the organizations that prepare or that distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, except for related associated detail specifications, specification sheets or MS standards, the text of this specification shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 General. Each projectile shall be properly marked and the surfaces either adequately prepared and primed or thoroughly cleaned and protected. Projectiles smaller than 75mm shall be marked as specified on the applicable drawing. Projectiles of 75mm and larger shall be marked in accordance with this specification, except as may be qualified by the applicable drawings.

3.3 Marking. The marking shall be stamped or otherwise suitably indented into the proper surfaces. The marking shall be legible and arranged and located in an orderly manner. The surfaces of empty projectiles on which marking may be required by this specification are the base plug, rotating band, body and windshield. The marking to be stamped on these surfaces is given in table 1. Typical arrangements of marking for various types of projectiles are shown on figures 1 and 2. Where an identification mark is specified for any piece (see 6.2), MIL-STD-130 and MIL-P-15024 shall apply. Arabic numerals and upper case letters shall be used. The size of characters used in marking shall be appropriate for the space available. In general, 75mm through 4 inch projectiles shall be marked with characters 0.125 inch high, 5-inch projectiles with characters 0.1875 inch high, and larger than 5-inch projectiles with characters 0.25 inch high. Periods after abbreviations may be

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

omitted, but decimal points in fractional quantities and hyphens in compound words or abbreviations shall be shown.

TABLE 1. Marking of projectiles.

Marking	Example	Remarks (see)
Base Plug, if any Automated Data List (ADL) Number Serial Number/Zone Symbol	 ADL 53711-2838596B (if required) 123/X	Figure 2
Rotating Band Automated Data List (ADL) Number Ammunition Lot Number	 ADL 10001-2838466-A XYZ85A001-013	Figures 1 & 2 3.2.1

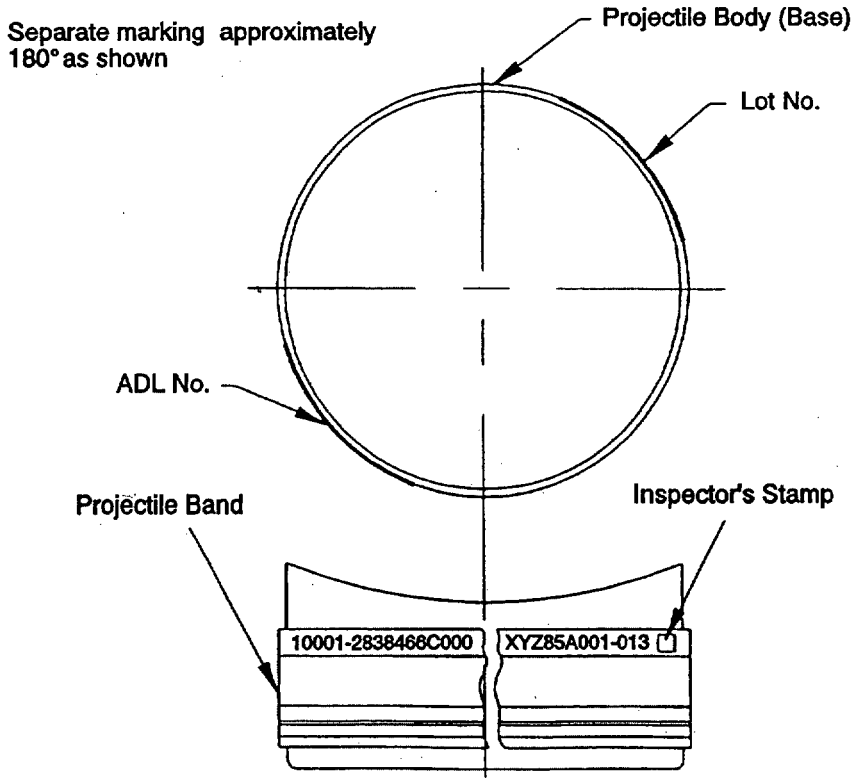


FIGURE 1. Typical arrangements of marking of projectiles without base plug.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

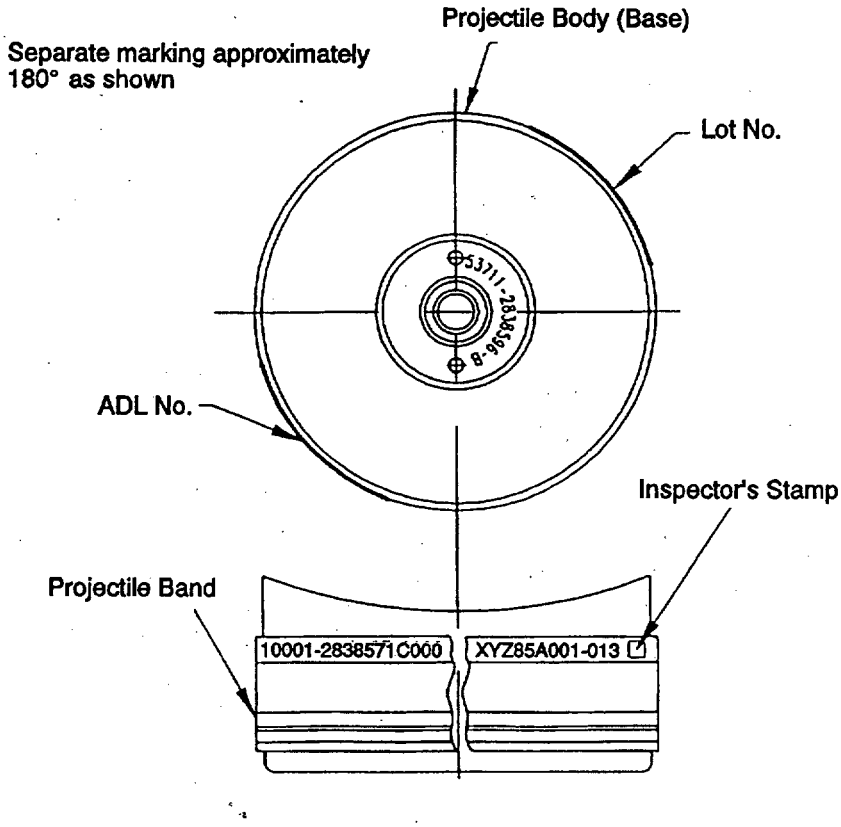


FIGURE 2. Typical arrangements of marking of projectiles with base plug.

3.3.1 Inspector's stamp. A block outline set apart from the rest of the marking shall be provided for the inspector's stamp (see figures 1 and 2), if required by the contracting activity (see 6.2).

3.3.2 Workmanship. Burrs raised by stamping shall be removed, as failure to gage after stamping shall be cause for rejection.

3.3.3 Ammunition lot number. The ammunition lot number shall be in accordance with MIL-STD-1168.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

3.3.4 Serial numbers. Serial numbers may be combinations of figures and letters at the option of the contractor. No two projectiles in the same lot shall bear the same serial number. Any base plug, windshield or other component except the rotating band, which is not entirely interchangeable with every other similar component in any lot, shall bear the same serial number as the projectile body (see 3.2.5). Unless otherwise specified (see 6.2), serial numbers are not required.

3.3.5 Zone symbol. For those components having press fit assembly after loading operations, a zone symbol may be used in lieu of serial numbers. Any component of the lot bearing a particular zone symbol shall provide an assembly within the limits specified on the applicable drawings when assembled with any of the equivalent number of matched components bearing the same zone symbol.

3.4 Grade A protection.

3.4.1 Surface preparation. When grade A protection is specified in the contract or purchase order (see 6.2), all interior surfaces of projectiles and removable components shall be thoroughly cleaned so as to be free of chips, dirt, scale, rust, grease, or other residues foreign to their composition; and after cleaning, there shall be no evidence of contamination from residual cleaning compounds.

3.4.2 Phosphate coating. After cleaning, all ferrous metal surfaces of projectiles (except the band seat area) and removable components shall be evenly coated with an adherent crystalline phosphate film in accordance with the requirements of TT-C-490, Type I, Class 1 or Class 2A. Coating weights shall be as specified in TT-C-490 unless otherwise specified (see 6.2). All the treated surfaces shall be thoroughly dry and clean without any contamination when the primer and other coatings are subsequently applied to the untreated phosphated surfaces.

3.4.3 Exterior surface primer application. The exterior surfaces of the projectile body (including threads on nose for fuze protecting cap), windshield, and nose excepting the rotating band, threaded surfaces, seating surfaces, and surfaces designated by applicable drawings, shall be uniformly covered with a primer. The primer shall conform to the revision of TT-P-664 specified in the contract or order (see 6.2), or MIL-P-22332. Care shall be exercised in controlling the limits of primed areas in order to prevent the impingement of paint on any surface which is to remain unprimed. The primer film shall be hard, tough, adherent, smooth, continuous and unbroken, and shall be thoroughly dry so as not to be damaged in any way at the time of packaging for delivery.

NOTE: If revision A, B, or C of TT-P-664 is specified in the contract or order, the projectiles shall be air dried for a minimum of 72 hours prior to being tested for dry film thickness, adhesion, or salt spray resistance.

If revision D of TT-P-664 is specified in the contract or purchase order, the projectiles should be heated to, and maintained at, a metal temperature of 150 degrees Fahrenheit (°F) for 30 to 45 minutes prior to being air dried for a minimum of 72 hours. Testing may then proceed in accordance with this specification. If the projectiles are not heated, they shall be air dried for a minimum of 144 hours prior to being tested for dry film thickness, adhesion, or salt spray resistance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

3.4.4 Interior surface primer application. The interior surfaces of projectiles and the applicable surfaces of base plugs, fuze adapters, and noses which will be in contact with explosives after the projectile is loaded, shall be uniformly coated with a smooth, continuous, and unbroken surface of acid-proof black paint (cavity paint) conforming to MIL-C-450 Type I or II. To aid in preventing runs or sags in the paint, leave the closure plug out of the nose threads for a minimum of 30 minutes or until the paint has lost its tackiness. The interior surfaces of windshields excepting threads shall also be painted.

3.4.4.1 Explosive compatibility. The interior surfaces of projectiles and the applicable surfaces of base plugs, fuze adapters, and noses which will be in contact with PBX type explosives after the projectile is loaded, and the cavities of load dispensing projectiles (CHAFF, Illuminating, Improved Conventional Munitions, etc.) shall be coated with a smooth, continuous, and unbroken surface of primer, MIL-P-22332 or TT-P-664, that meets the following reactivity requirements:

The primer shall not react with plastic bonded, cast explosive, and the maximum volume of reactivity gas liberated shall be 2 milliliters per gram (ml/g) when tested in accordance with 4.6.3.

3.4.5 Coating of unprotected surfaces. Threads and seating surfaces of projectiles, base plugs, nose adapters, fuze hole closures, and windshields shall be given a thin, even coat of a suitable nonhardening, rust inhibiting grease. The grease shall conform to MIL-G-81322 and shall be applied in such a manner so as to prevent oxidation of unprimed metal without contaminating the primed surfaces.

3.4.6 NO-GO gaging of threaded surfaces. Threaded surfaces shall comply with NO-GO thread gaging requirements before application of phosphate coating and shall comply with all thread gaging requirements after application of phosphate coating.

3.5 Grade B protection.

3.5.1 Surface preparation. When grade B protection is specified in the contract or purchase order (see 6.2), interior and exterior surfaces of the projectiles and removable components shall be thoroughly cleaned so as to be free of chips, dirt, scale, rust, grease, or other residues foreign to their composition, and after cleaning, there shall be no evidence of contamination from residual cleaning compounds. Objectionable dirt residues include perspiration, residual welding or brazing fluxes, cutting or cooling compounds, buffing or grinding compounds, residual heat-treat salts, general factory contamination from machines, floors, etc., dust and chemical deposits, and residues from etching solutions. Cleaning shall be carefully performed in strict accordance with the requirements of this specification. The applicable detail requirements for Type II of TT-C-490 for cleanliness of surfaces shall be for all metallic surfaces. All surfaces shall be thoroughly dry when coated with the preservative.

3.5.2 Preservative application. All interior and exterior surfaces of the finished and cleaned projectiles and removable components shall be thoroughly and evenly coated with a rust preventive compound. Unless otherwise specified in the contract or purchase order (see 6.2), the rust preventive compound shall comply with the requirements of Grade 1 of MIL-C-16173. The application of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

preservative should follow the cleaning without delay, allowing only enough time for cooling of any heated parts to the proper temperature for preservative application. A thin, even coat of grease conforming to MIL-G-81322 shall be applied to all internal and external threads prior to the overall application of the grade 1 compound specified above.

3.5.3 Method of preservative application. The preservative, at a temperature of 70 to 95°F, shall be applied to a film thickness of not less than 0.002 inch by any suitable method such as dipping, spraying, slushing, or brushing which will produce a continuous, unbroken, even and uniform coating free of moisture droplets, air bubbles, or other inclusions. Excess preservative shall be drained from surfaces in such a manner as to avoid accumulation of the compound in cavities, apertures, and corners of the projectiles and components. During the entire period of preservative application, the projectiles and components shall be suitably supported or racked so that the protective film is undisturbed until dry and sufficiently set for handling and packaging. Coated parts which are subsequently heated to dry to set the preservative compound shall be allowed to cool before handling. Just prior to packaging, any bare or uncoated spots which result from contact with supports or racks during the coating and drying procedure shall be touched up by brushing with additional preservative.

3.6 Dry film thickness. The thickness of the dry film coating (phosphate and primer) shall be within the limits specified in 3.6.1 through 3.6.3.

3.6.1 Exterior. Projectiles shall have a phosphate and primer coating of 0.0005 to 0.0012 inch thickness on all primed exterior surfaces.

3.6.2 Bourrelets. Projectile bourrelets shall have a phosphate and primer coating of 0.0005 to 0.0010 inch dry film thickness. If electromagnetic thickness gage readings exceed 0.0010 inch at the bourrelet, when tested in accordance with 4.6.4, a ring gage of the diameter specified in the contract or purchase order (see 6.2) shall pass over the bourrelet for the projectile to be considered acceptable.

3.6.3 Interior. The minimum interior phosphate and primer dry film thickness shall be 0.0025 inch when tested in accordance with 4.6.4. Exceptions to the minimum thickness shall be as specified on the particular projectile body drawing.

3.7 Adhesion. Projectile bodies shall show satisfactory primer and phosphate adhesion when tested in accordance with 4.6.5. Unsatisfactory adhesion is indicated by the following conditions found under the removed tape:

a. Exposure of bare metal indicating failure of both the phosphate and primer to adhere properly.

b. Exposure of the phosphate coating in any one area exceeding 0.125 (1/8) inch average diameter.

c. Exposure of the phosphate coating in more than one area exceeding 0.0625 (1/16) inch average diameter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

d. Exposure of the phosphate coating in more than five areas less than 0.0625 (1/16) inch diameter.

3.8 Salt spray. Projectile bodies shall not exhibit any of the following conditions after being subjected to the salt spray test of 4.6.6:

a. Evidence of primer film failure such as blistering, softening, or loss of adhesion of the primer more than 0.0625 (1/16) inch on either side of the scribe marks.

b. At all other points, no more than a trace of primer film failure, and not more than five scattered blisters, none larger than 0.0469 (3/64) inch in diameter, on an area of 48 square inches. On an area less than 48 square inches, a proportionally smaller number of failed areas are permitted.

c. After removal of the primer from the scribe mark area, no rusting, pitting, or rust creepage exceeding 0.125 (1/8) inch on either side of the scribe marks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Test samples. As specified in the contract or purchase order (see 6.2), test samples shall be prepared from actual production items or portions thereof, or from scrap items of the same kind and finish which have been rejected for causes other than phosphating, material composition, or heat

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

treatment. If a portion of an item is used, it shall have at least 48 square inches of surface area on at least one surface.

4.3.1 Test panel substitution. Substitution of test panels in lieu of actual production items for the tests of 4.6 must be approved by the Contracting Officer's Technical Representative. If the use of the test panels is approved, the panels shall be prepared with the same surface finish and shall have been phosphated and primed during the same operations that coated the lot of projectiles the panels represent.

4.4 First article inspection. A minimum of three samples shall be prepared for first article testing (see 6.2). The samples shall have been phosphated and primed on the same production line as will be used for regular production and allowed to dry in accordance with 3.4.3 and 3.4.4. Production shall not proceed until the first article samples have successfully met the requirements of 3.6 through 3.8 except at the contractor's risk.

4.5 Quality conformance inspection. Unless otherwise specified in the contract or purchase order (see 6.2), three projectile bodies shall be randomly selected from each production lot and subjected to the tests of 4.6.4 through 4.6.6. After two consecutive production lots have successfully met all the requirements of 3.6 through 3.8, sampling may be reduced to two rounds per production lot. After two more consecutive production lots have successfully met all the requirements of 3.6 through 3.8, sampling may be reduced to one projectile per production lot. If any sample fails to meet all the requirements of 3.6 through 3.8, or the cleaning, phosphating or priming processes, or materials are changed, or production has been discontinued for more than 90 days, sampling shall revert to three projectiles per production lot. Reduced inspection may then be granted as in the above sequence. If a failure to meet the requirements of 3.6 through 3.8 occurs, all of the projectiles processed since the last accepted production lot shall be rejected and corrective action taken.

4.6 Tests.

4.6.1 Testing sequence. Sample testing shall be performed in the following sequence: visual inspection, explosive compatibility, dry film thickness, adhesion, and salt spray.

4.6.2 Visual inspection. Projectiles shall be examined visually and manually for conformance with 3.3 through 3.5. Failure to meet these requirements shall be cause for rejection.

4.6.3 Explosive compatibility test. The samples shall be tested for conformance to 3.4.4.1 in accordance with Method 504.1.1 of MIL-STD-650 with the following exceptions:

a. The test sample shall consist of 0.2 gram of pulverized explosive plus 0.2 gram of test material.

b. Heat sample for 48 hours at 100°C.

c. The length of the top leg of the tube may be 15 centimeters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

d. Gassing volumes (A, B, and C) shall be calculated as ml/g for each test tube sample. Change the equation, paragraph 5.5 of Method 504.1.1 to read as follows:

$$\text{Gas Due to Reactivity, ml/g} = C - \frac{(A + B)}{2}$$

After a lot of primer has been determined to be compatible with the explosive, testing is not required for subsequent lots provided that certification can be obtained from the manufacturer that the component materials and manufacturing process were not changed. If certification cannot be obtained from the manufacturer and primer lots change, or primer manufacturer is changed, reactivity shall be redetermined.

4.6.4 Dry film thickness. The dry film thickness (phosphate and primer) shall be measured by an electromagnetic thickness gage for conformance to 3.6. The instrument shall have been standardized on the same type of surface finish as that over which the film has been applied. The maximum indicated exterior dry film thickness shall not be more than 0.0015 inches. The minimum indicated interior dry film thickness shall not be less than 0.0022 inches. These maximum/minimum instrument readings allow for gage tolerance and are not intended to relieve the contractor from meeting the requirements of 3.6. A minimum of three measurements at least 3.0 inches apart shall be taken on each sample. Any measurement failing the thickness requirements of 3.6 shall fail the lot.

4.6.5 Adhesion. Surfaces shall be tested for coating adhesion in the following manner:

a. A piece of 3/4 inch wide, pressure-sensitive adhesive tape conforming to commercial item description A-A-1830 shall be pressed firmly onto a flat or cylindrical surface of the sample. At least two inches of tape shall be in contact with the sample. All air bubbles should be removed from under the tape by rubbing its surface. Wait approximately 10 seconds before proceeding, to allow the test area to return to ambient temperature.

b. Grasp a free end of the tape and rapidly strip it from the sample by pulling the tape back upon itself in such a manner that the tape is folded back during the procedure.

c. Observe the area under the tape for removal of primer/phosphate coating.

Failure of the sample to meet the requirements of 3.7 fails the lot.

4.6.6 Salt spray. Test the samples for salt spray resistance in the following manner:

a. Scribe two perpendicular lines approximately two inches in length into the surface of the samples with a sharp instrument, sufficiently deep to expose bare metal. Care should be taken to ensure exposure of bare metal with one pass of the scribe for each line.

b. The prepared sample shall then be exposed to the salt spray test of ASTM B117 for 144 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

c. Remove the sample from the salt spray apparatus and examine for evidence of primer film failure in accordance with requirements of 3.8.

d. The primer shall then be carefully removed from the area of the scribe marks with paint remover so as not to disturb the underlying phosphate coating, and the resistance of the phosphate coating to the salt spray examined in accordance with 3.8.

Failure of the primer or phosphate coatings on any one sample to meet the requirements of 3.8 shall fail the lot.

5. PACKAGING

5.1 Packing and marking. Preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of the applicable projectile specification or drawing.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended Use. This specification addresses the protection and marking of projectiles and is to be used in conjunction with the applicable projectile manufacturing document.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Applicable projectile manufacturing document package (see 1.1)

c. Grade of protection required (see 1.2 and 3.4, and 3.5)

d. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1)

e. Whether a first article sample is required (see 3.1)

f. Identification mark required for any piece (see 3.3)

g. Whether inspector's stamp is required (see 3.3.1)

h. Whether serial numbers are required (see 3.3.4)

i. If phosphate coatings weights are to be other than as specified in TT-C-490 (see 3.4.2)

j. The revision of TT-P-664 which covers the desired primer (see 3.4.3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

k. Type or grade of rust preventive compound, if other than as specified (see 3.5.2)

l. The diameter of ring gage required (see 3.6.2)

m. The agency to perform inspection requirements if other than as specified (see 4.1)

n. Sample plans and procedures to be used (see 4.3, 4.4, and 4.5).

6.3 First article. When a first article inspection is required, the sample should be selected from the first projectiles prepared and conditioned against this specification. The first article should consist of three or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Grade A protection. Grade A protection is recommended when extreme climate exposure is anticipated (see 3.4).

6.5 Grade B protection. Grade B protection is intended to preserve the projectiles and components against corrosion for periods up to 1 year between the time of manufacture and loading at ammunition activities provided undercover storage is maintained. If prolonged outdoor storage without cover is anticipated, rust preventive compound conforming to MIL-C-18487, which is more difficult to remove, should be specified and applied to a minimum film thickness of 0.003 inch in lieu of the compound and thickness specified in 3.5.

6.6 Subject term (key word listing).

Corrosion, projectile
Preservation, projectile
Protection, surface
Rust prevention, projectile

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy-OS
(Project 1320-NB03)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-18948B(OS)	92 JULY 15

3. DOCUMENT TITLE
PROJECTILES, PRIMING, MARKING, AND GREASING OF

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER	(1) Commercial 301-743-4358/4510	(2) AUTOVON 364-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVE. INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:08Z
Check the source to verify that this is the current version before use.